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# Treatment of Acromegaly, Cushing Disease and Nelson Syndrome

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*Since 1957 we have treated more than 429 patients who had pituitary neoplasms, most of which were hormone-secreting tumors. Long-term follow-up in the large group of patients treated for acromegaly shows a median survival of better than 16 years, with improvement over time. The short-term follow-up results in patients with Cushing* disease, Nelson syndrome and chromophobe adenoma are very encouraging. To compare these excellent results with those following surgical procedures, a large study of patients followed for a long period after the operations is needed.*

SINCE 1957 we have had an intensive experience in the study, diagnosis and treatment of acromegaly, Cushing* disease and Nelson syndrome. This report emphasizes the long-term results chiefly in treating patients with acromegaly. Previous reports have given extensive information regarding the methods used, the physical aspects and dosimetry of the radiation and the many detailed metabolic studies.[1-12] In the past and present as well, these disorders have been and are treated by surgical procedures or by conventional pituitary irradiation, or both. However, in our work, heavy-particle irradiation (plateau or Bragg Peak) has been used,[13] administered with the aid of a stereotactic setup to hold the patient's head firmly, and by head and body rotation, to limit to safe levels the amount of radiation delivered to the cranial nerves and temporal lobes.

During the "Cushing era" patients with Cushing disease were treated with standard roentgen irradiation[14-17] The treatment was first carried out by Dr. Merrill Sosman, a radiologist at the Peter Bent Brigham Hospital in Boston. However, the doses given were insufficient.[17] The first patient receiving this treatment at Peter Bent Brigham Hospital was diagnosed as having a large basophilic adenoma. Several years later the patient died and a postmortem examination showed a large basophilic adenoma of the pituitary. In the intervening years many patients with Cushing disease have been treated by bilateral adrenalectomy. However, in recent years it has become recognized that the primary defect is either in the pituitary gland itself or in the hypothalamopituitary adenal axis, and the point of attack should be the pituitary gland. More recently, with the "rediscovery" of transsphenoidal pituitary operations, a greater number of patients with Cushing disease are being treated by hypophysectomy, with removal of either microadenomas or macroadenomas.[18] However, there has not been adequate follow-up time, especially for those with acromegaly, in any large series of such patients to allow analysis of long-term effects regarding the return to normal of metabolic findings and the relief of clinical symp-

*THE WESTERN JOURNAL's style regarding eponyms is that they are not written in the possessive form; therefore, Graves disease, Ewing sarcoma and Paget disease. A explanation may be found on page 78 of the July 1978 issue.

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toms and signs. It will be another five or more years before such a series of patients will have been followed long enough for comparison with our series.

## Patients and Methods

We have treated 429 patients with alpha-particle pituitary irradiation (APPI, also referred to as heavy-particle pituitary irradiation [HPPI]) since 1957. These patients had pituitary tumors, most of which caused hormonal hypersecretion. As stated above, in treating these patients we used a beam of alpha particles or protons, employing a stereotactic method for alignment of the pituitary sella and delivering to the pituitary gland doses close to twice that delivered by the usual photon method (Figure 1). In this way, we have observed some remarkable results. The method consists of rotation of the patient's head right and left through angles of 30 degrees and by simultaneous rotation of the body. Thus, one is able to deliver to the pituitary gland a much larger dose than is possible with photons, without damage to the nervous tissue such as the cranial nerves and the temporal lobes.

In computerized settings of the dosage schedule for each patient, it is calculated that less than 3,000 rads be directed to the temporal lobe and, on the average, 6,000 rads are given over five days. In some patients, especially in the early cases, we used the so-called Bragg peak,[19] as has Kjellberg,[20] who also has had extensive experience with Bragg peak protons. We also treated a few patients who had received previous photon irradiation, but since 1961 this has been discontinued and we no longer treat any patients who have had previous radiation therapy. We no longer use the Bragg peak because of the difficulty of exact alignment. Recent work, however, with another new beam of much heavier particles such as carbon 12 or neon, and a method of so-called autoradioactive localization for the location of the peak, should make it possible to give larger doses than are given now by the direct beam plus head and body rotation.[4] This will be especially important in the case of Cushing disease where, at present, some patients should be given medical treatment with metyrapone or aminoglutethamide, or both, for several months until the effects of radiation have taken place. Some of these patients

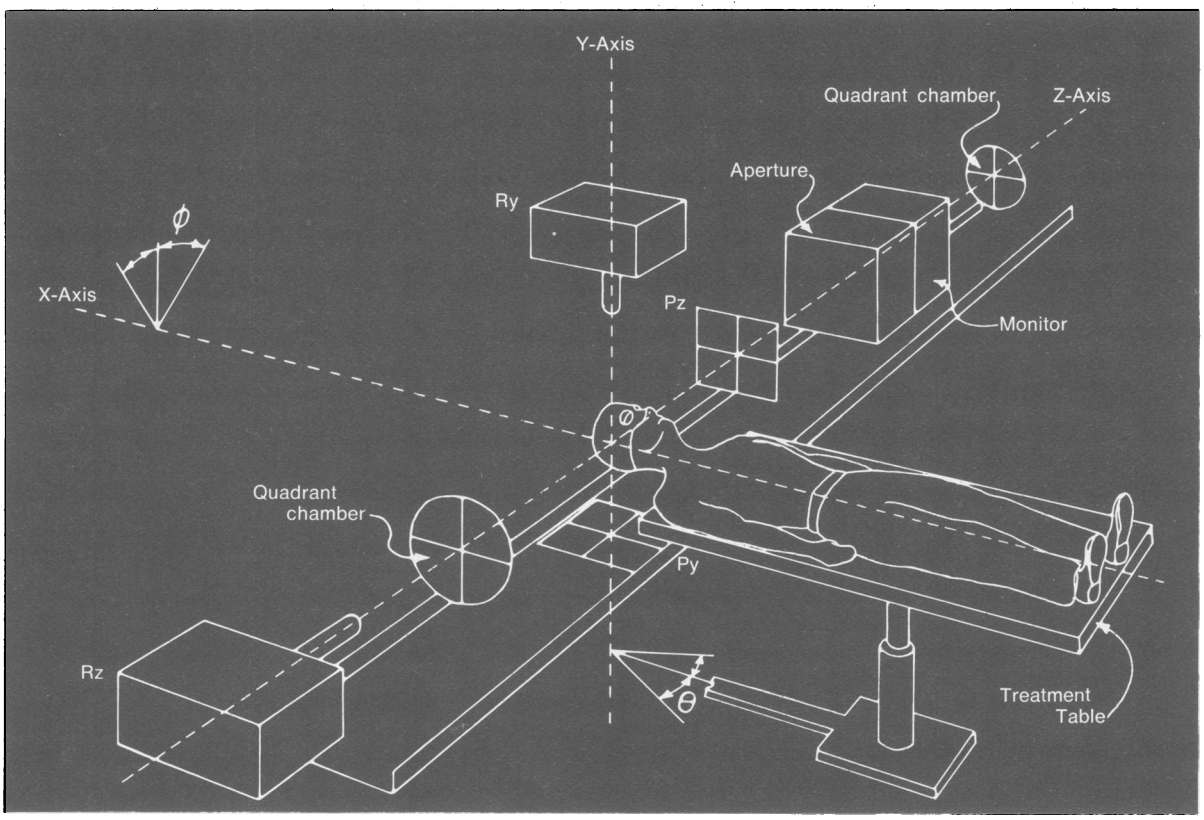



**Figure 1.**—Diagram of the stereotactic setup for administration of heavy-particle irradiation to the sella turcica. Reproduced by permission from Am J Roentgenol Radium Ther Nucl Med 67:2, 1952.

require higher doses of radiation than we can now give. We have found by using all available methods for visualization of the sella turcica that the most reliable are pneumoencephalography and computerized tomographic (CT) scans. In all patients who are treated by the ion beam methods such studies are carried out to avoid treating patients with suprasellar extensions or massive sphenoid extension. However, we do treat patients who have sphenoid extensions in cases where we can appropriately place the beam. Although visual field defects are usually associated with suprasellar extension, we have treated a few patients who had very small defects. The goals of the treatment are to relieve or minimize symptoms and signs and to try to return metabolic findings to normal (Figure 2).

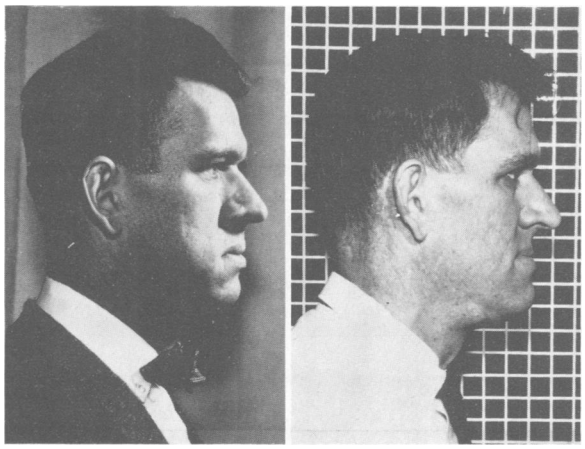


**Figure 2.**—Patient before treatment for acromegaly and ten years later. After treatment soft tissue regression has taken place and human growth hormone levels became normal. Reproduced by permission from J Clin Endo Crinol Metab 31:188, 1970.

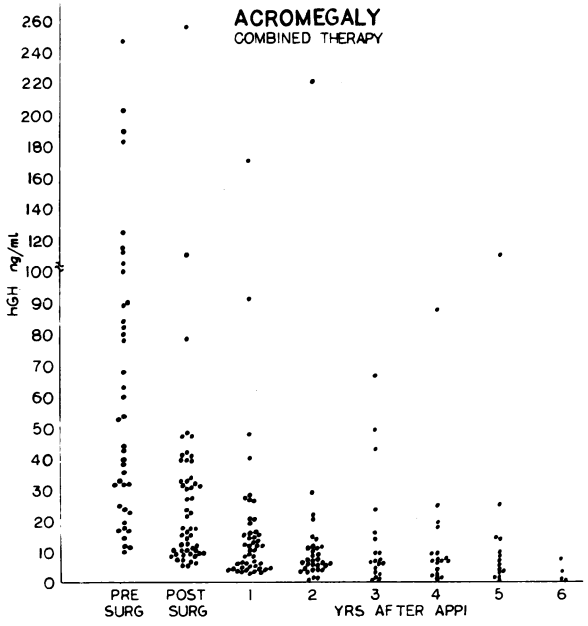



**Figure 3.**—Human growth hormone (hGH) values for patients before and after unsuccessful surgical treatment, and after alpha-particle pituitary irradiation (APPI, also referred to as heavy-particle pituitary irradiation [HPPI]). Reproduced with permission from Linfoot JA (Ed): Recent Advances in the Diagnosis and Treatment of Pituitary Tumors, Raven Press, New York, 1979.

*Acromegaly*

Many of the patients with acromegaly had had previous unsuccessful operations, usually transsphenoidal surgical excisions. Figure 3 shows human growth hormone (HGH) values in a group of patients before and after these unsuccessful operations, and after heavy-particle pituitary irradiation (HPPI). Figure 4 shows growth hormone levels in all of our patients with acromegaly before and after HPPI. Figure 5 compares survival of patients with acromegaly after therapy (median 16 years) with survival in a diabetic population and expected survival in an age-matched population. It should be mentioned that as a group our patients have often been refused surgical treatment for various reasons. In general, our patients are selected if the beam can be placed accurately. If not, we refer such patients for surgical treat-

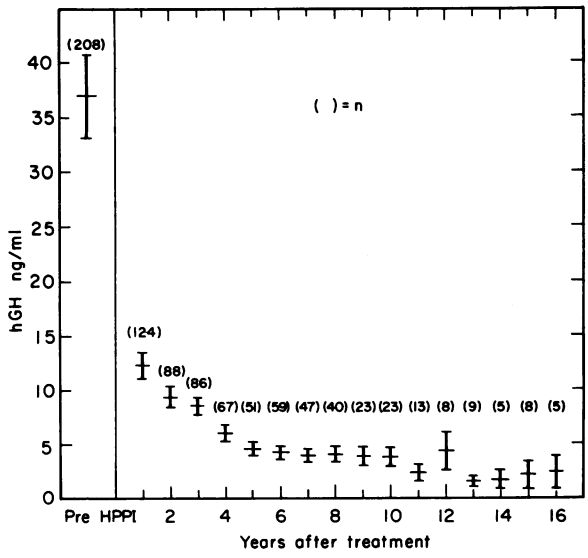



**Figure 4.**—Human growth hormone (hGH) values in patients with acromegaly before and after heavy-particle pituitary irradiation (HPPI). Reproduced with permission from Linfoot JA (Ed): Recent Advances in the Diagnosis and Treatment of Pituitary Tumors, Raven Press, New York, 1979.

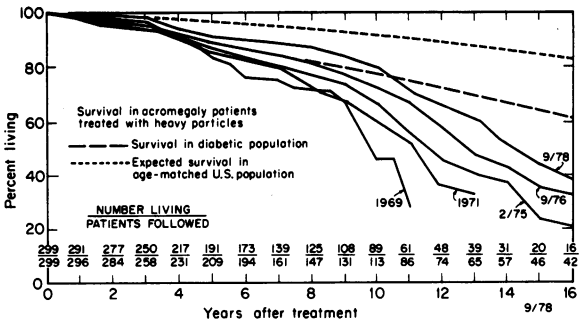



**Figure 5.**—Survival of patients with acromegaly after heavy-particle pituitary irradiation compared with survival in a diabetic population and expected survival in an age-matched population. Reproduced with permission from Linfoot JA (Ed): Recent Advances in the Diagnosis and Treatment of Pituitary Tumors, Raven Press, New York, 1979.

ment. In the early part of our work, before 1961, we had patients with rare neurological complications; however, since then less than 1 percent of the patients have had any such complications. Table 1 indicates the age and sex distribution of patients with acromegaly and the delay before patients are referred for treatment.

## Cushing Disease

There are 64 patients in this group; of these, 59 had normal sellar sizes. The diagnosis was made by the usual clinical and metabolic findings, including the dexamethasone suppression test and a metyrapone test. After therapy, replacement therapy with cortisone was necessary in 9 percent of these patients, thyroid replacement in 16 percent and gonadal replacement therapy in 11 percent. There were two deaths: one patient died of cerebrovascular complications unrelated to the Cushing disease, which was under control. The other was a patient in a mental hospital in whom Cushing disease was also under control. There were no other complications. This series is being increased in number, and longer years of follow-up will be necessary for comparison with long-term results in patients treated surgically.[18]

Figure 6 shows a patient with Cushing syndrome before and after therapy. After treatment cortisol levels and most of the symptoms returned to normal.

### *Nelson Syndrome*

By Nelson syndrome we mean what Nelson has termed the third stage of Cushing syndrome,[21] and which he describes as having the following characteristics:

A third stage (most common after total adrenalectomy) in which plasma ACTH levels are often at extremely high levels, the patients are intensely pigmented and the plasma ACTH is poorly suppressed by the administration of exogenous cortical steroid. This group of patients is most often found to have pituitary tumors, although such tumors have been described in the earlier stages of the disease.

According to reports of several authors,[21-28] pituitary tumors have developed in about 10 percent of patients with Cushing disease associated with adrenocortical hyperplasia who had been treated initially with total adrenalectomy. These authors expressed concern that adrenalectomy may serve as a triggering mechanism (lack of feedback) that is responsible for the clinical ap-

TABLE 1.—*Age and Sex Distribution of 294 Patients With Acromegaly Treated With Heavy-Particle Pituitary Irradiation*

| Patients | | Age (years) | | Duration (years) |
|---|---|---|---|---|
| Sex | Number | At Time of Heavy-Particle Therapy Median (Range) | At Time of Onset of Signs and Symptoms Median (Range) | At Time of Heavy-Particle Therapy Median (Range) |
| Male | 181 | 39.8 (15-69) | 31.2 (13-59) | 7.2 (1-40) |
| Female | 113 | 48.0 (15-68) | 36.8 (13-63) | 8.4 (1-29) |
| TOTAL | 294 | 43.5 (15-69) | 34.5 (13-63) | 7.7 (1-40) |

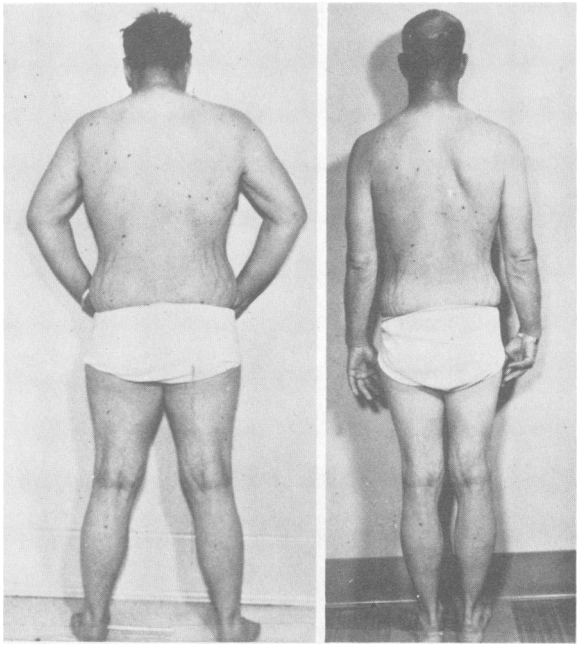


**Figure 6.**—Patient with Cushing disease before and after therapy. After treatment there is a loss of the Cushinoid features, such as striae and obesity. Reproduced by permission from Transactions of the American Clinical and Climatological Association, Vol. 81, 1970.

pearance and rapid growth of the pituitary tumors. Orth and Liddle[25] reported that in 19 patients treated first by pituitary irradiation and later by bilateral adrenalectomy, no evidence of pituitary tumors was found during a follow-up period of 1 to 13 years. They pointed out the possibility that in patients who had received pituitary irradiation before total adrenalectomy large pituitary tumors are less likely to develop. However, Wild, Nicolis and Gabrilove[28] cited two of their patients who initially had undergone unilateral adrenalectomy and pituitary irradiation (conventional radiotherapy). These patients subsequently required total adrenalectomy to control their disease and, later, skin pigmentation and pituitary tumors developed. The authors concluded that experience to date indicates that pituitary irradiation may reduce the incidence of Nelson syndrome, but that its effectiveness had not been precisely established. We also have observed Nelson syndrome in two instances after treatment of patients with Cushing disease.

There have been 15 patients in the study with Nelson syndrome. Six had unsuccessful surgical treatment before receiving HPPI. There has been one death due to an invasive tumor, which is not uncommon in patients with tumors that secrete adrenocorticotropic hormone (ACTH). There were no other complications, and the pigmentation of the skin either disappeared or decreased in all patients treated.

### *Prolactin Tumors*

We have had limited experience in treating patients with prolactin-secreting tumors. Of the 29 treated, nine had previous unsuccessful hypophysectomies. Five patients had sellar volumes of less than 1,100 cu mm in size and thus were considered to have microadenomas. The other patients had grade II, III or IV lesions, with relatively large sella turcicas. About 10 percent of these patients required adrenal cortisone replacement therapy. However, in general, the patients have had a lowering of prolactin levels and disap-

TABLE 2.—*Patients With Pituitary Tumors Treated With Heavy-Particle Pituitary Irradiation*

| | Number of Patients | | | Survival, Years After Heavy-Particle Therapy | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 1 | | 1-2 | | 3-4 | | 5-9 | | 10-14 | | 15-19 | |
| | Total | Living | Dead | Living | Dead | Living | Dead | Living | Dead | Living | Dead | Living | Dead | Living | Dead |
| Acromegaly | 294 | 268 | 26 | 10 | 5 | 53 | 3 | 37 | 10 | 110 | 3 | 44 | 3 | 14 | 2 |
| Cushing disease | 54 | 52 | 2 | 5 | 1 | 19 | 0 | 3 | 0 | 15 | 1 | 9 | 0 | 1 | 0 |
| Nelson syndrome | 14 | 13 | 1 | 1 | 0 | 3 | 0 | 2 | 1 | 4 | 0 | 2 | 0 | 1 | 0 |
| Chromophobe adenoma | 32 | 27 | 5 | 2 | 0 | 4 | 2 | 4 | 2 | 10 | 1 | 5 | 0 | 2 | 0 |
| Prolactin-secreting adenoma | 16 | 16 | 0 | 4 | 0 | 5 | 0 | 4 | 0 | 2 | 0 | 1 | 0 | 0 | 0 |
| TOTAL | 410 | 376 | 34 | 22 | 6 | 84 | 5 | 50 | 13 | 141 | 5 | 61 | 3 | 18 | 2 |

TABLE 3.—*Cyclotrons and Synchrotrons Capable of Producing Heavy Particles With Sufficient Energy for Therapeutic Use*

| Location of Cyclotron or Synchrotron | Proton Energy |
|---|---|
| University of Maryland, College Park, Maryland | 100 MeV |
| Harvard University, Cambridge, Massachusetts | 160 MeV |
| University of Rochester, Rochester, New York | 240 MeV |
| Columbia University, Irvington, New York | 385 MeV |
| Carnegie Institute of Technology, Pittsburgh, Pennsylvania | 440 MeV |
| University of Chicago, Chicago, Illinois | 450 MeV |
| NASA-Langley Research Center, Langley Field, Virginia | 600 MeV |
| Lawrence Berkeley Laboratory (184"), Berkeley, California | 730 MeV |
| Princeton University, Princeton, New Jersey | 3 BeV |
| Lawrence Berkeley Laboratory (Bevatron), Berkeley, California | 9 BeV |
| McGill University, Montreal, Quebec, Canada | 100 MeV |
| College de France, Orsay, France | 155 MeV |
| Atomic Energy Research Establishment, Harwell, England | 175 MeV |
| University of Uppsala, Uppsala, Sweden | 185 MeV |
| Institute for Theoretical and Experimental Physics (ITEP), Moscow, U.S.S.R. | 200 MeV |
| University of Liverpool, Liverpool, England | 381 MeV |
| European Council for Nuclear Research (CERN), Geneva, Switzerland | 600 MeV |
| Joint Institute for Nuclear Research (JINR), Dubna, U.S.S.R. | 680 MeV |

pearance of galactorrhea and amenorrhea; and in at least two patients pregnancy has occurred. At present, our experience is not extensive enough for a detailed discussion of the long-term effects of this therapy in these patients.

## Conclusions

The most important message of this paper is that to compare the various methods of therapy for pituitary tumors, it is essential that hundreds of patients be followed for many years after treatment. It will require a good many years before the long-term results of surgical treatment of acromegaly can be compared with our results. We emphasize that in the case of acromegaly, the median survival of patients is better than 16 years, and this will improve over time (Figure 5).

It also should be emphasized that patients are not refused our treatment program although many of them had been previously turned away as poor surgical risks elsewhere. As a group, our cases are more complicated than those patients receiving surgical treatment. Table 1 shows that patients with acromegaly are treated approximately eight years after the diagnosis has been made. If the diagnosis is made sooner and treatment administered earlier, it seems that survival may well approach normal.

Death in patients with acromegaly is primarily due to vascular disease. Also, approximately a third of patients with acromegaly have hypertension and about half of these have a return of blood pressure to normal or in the direction of normal after irradiation therapy. The mechanisms of hypertension in these patients are under intensive study in our laboratory, but they are not understood yet. In patients with advanced acromegaly and cardiomegaly (of which we have six) the heart does not return to normal size.

Finally, Table 2 provides a summary of 410 of the patients treated between 1957 and September 1978, and Table 3 lists cyclotrons and synchrotrons which are being used or could be used for heavy-particle pituitary irradiation as described in this paper.

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